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Summary

AI Research Engineer & Software Architect with 14+ years building enterprise platforms and production AI systems. Currently cost & efficiency lead for Cisco's **Circuit/MyAgent** enterprise AI platform — model routing and cascading, OSS-model adoption, and the agent-evaluation infrastructure behind model selection. Previously architected AssessmentX AI, a healthcare AI platform deployed across 40+ hospital systems and 400+ assessment projects. Combines agent evaluation, hybrid retrieval, multi-agent orchestration, MCP, RAFT/LoRA fine-tuning, and AI observability with production architecture across Python/FastAPI, Postgres, Java/Spring, and cloud-native systems. **M.S. Computer Science (AI), Georgia Tech, 2024.**

Independent Research & Open-Source Systems

MAKBench — agent benchmarking & cost-observability harness with a published leaderboard

[🔗 Code](#) [🌐 Leaderboard](#)

- Hermetic per-attempt Docker sandboxes with an LLM gateway as the single source of cost truth (budget-capped virtual keys — metered, never self-reported), split-phase grading by a trusted verifier the agent never sees, and anti-triviality checks; live leaderboard covering **2 agents × 7 models × 10 enterprise tasks** with pass@1, pass^k reliability, and cost-of-success.

HCFT — healthcare-domain 12B RAFT fine-tune for grounded RAG and calibrated refusal

[🔗 Code](#) [🌐 Write-up](#)

- Trained a full-precision **bf16 LoRA** on 8×H200 GPUs (DDP) with RAFT data generated from a frozen hybrid retriever (dense + BM25 + RRF + rerank; hit@5 ~0.78); under a bias-controlled three-model-family judge panel the 12B **tied GPT-5.4-mini head-to-head** and led calibration at **79.5% balanced accuracy vs. 67.4% / 57.3%**, refusing **74% of unanswerable questions vs. 16%** for the full frontier model.

Laika — BYOK coding agent for VS Code (*in active development*)

[🔗 Code](#)

- Provider-agnostic coding agent (host-agnostic core runtime, VS Code extension host, React sidebar) with **tiered autonomy** (manual / guarded / autonomous) governed by a per-repo policy file, and “Orbit” — a live overview of the in-flight change set.

Conclave — multi-expert LLM deliberation rooms

[🔗 Code](#)

- Multi-expert deliberation where **every expert drafts its own solution from scratch, then proposes and critiques changes with rationale — the way developers review a PR — and the merged edits execute only once the room converges**; FastAPI + async SQLAlchemy with Postgres as system of record, job queue (FOR UPDATE SKIP LOCKED), pause signal, and event feed.

Experience

Cisco Systems

Nov 2018 – Present

San Francisco Bay Area, CA

AI Research Engineer | Cost & Efficiency Lead, Circuit/MyAgent Platform

Jul 2026 – Present

Technical leadership on Cisco's enterprise AI team, owning cost, efficiency, and model optimization — OSS and lower-cost model adoption via routing and cascading — across the Circuit/MyAgent platform (enterprise Q&A + the Cisco MyAgent agent).

- Agent evaluation program and platform.** Qualified a faster, lower-cost alternate model and secured MyAgent's on-schedule launch through a model-agnostic evaluation spanning 1,400+ workflows and live MCP tool use — **70,651 trials (828,206 agent steps)** across frontier and in-house models with fault injection, LLM-as-judge scoring, and per-trial quality/latency/cost telemetry — productionized as a team-scale platform (FastAPI, Postgres-backed job queue, React live monitoring, 51-chart Superset analytics).
- Continuous model qualification.** Leading controlled control-vs-experiment studies — a baseline agent paired against a Rust-based experimental target — that qualify candidate models across short- and long-running task workflows and extended conversations under identical prompts, tools, and scoring; the evidence base for the platform's routing, cascading, and deployment decisions.

Software Architect – AssessmentX AI Platform

Nov 2024 – Jun 2026

Hands-on architect for AssessmentX AI, Cisco's production platform for knowledge ingestion, structured and semantic search, grounded chat, and multi-skill agentic analysis across 40+ hospital systems and 400+ assessment projects. Defined shared architecture across agent orchestration, model routing, retrieval, evaluation, observability, governance, and developer tooling.

- Reusable agent platform.** Built a domain-agnostic LangGraph SDK powering three production agents through declarative skill manifests, five graph templates, three-tier model resolution, Pydantic-validated outputs, and bounded retries — replacing bespoke plan/act/validate implementations.
- Multi-intent retrieval and data agent.** Designed a schema-aware planner that routes each turn across grounded RAG, NL-to-MongoDB aggregation, or multi-step analysis. Combined vector and BM25 retrieval with RRF and BGE/LLM reranking; reduced the route/classify/plan chain from ~3 LLM calls to one, lowering per-turn token cost.
- Supervisor analysis runtime.** Architected parallel specialist fan-out with structured outputs, bounded recovery, and live SSE topology traces; reduced manual research by **40–50 hours per customer engagement**.
- Governed spec-to-code agent.** Built markdown-to-Python assessment generation with AST allowlisting, SHA-256 human approval, immutable snapshots, and memory/time-capped sandbox execution; cut delivery time for a new assessment type from **~4 weeks to ~25 minutes** and reduced code-generation cost through model routing.
- AI observability and production safety.** Created self-hosted HIPAA-aware DAG/span tracing, token and cost attribution, LLM-as-judge scoring, and run comparison; preserved data sovereignty, removed **~\$20K/year in cross-charges**, and enforced grounding, refusal, and tenant isolation.

- **Agentic SDLC framework.** Established ~55 reusable multi-IDE skills, an age-encrypted secret vault, a TypeScript MCP server (31 tools / 10 resources / 5 prompts), and CSDL/OWASP security gates for AI-native delivery.

Software Engineer IV – Healthcare Assessment Platform

Sep 2022 – Nov 2024

Built a healthcare assessment-automation platform that compressed weeks of solution-architect effort into minutes of repeatable ingestion, scoring, recommendations, and deliverable generation for hospitals, providers, and life-sciences customers.

- **Assessment automation platform.** Architected and shipped the platform end-to-end across Java 21 / Spring Boot 3.5 microservices, a Python AI layer, and Angular 18; automated specification ingestion, third-party HIMSS scoring intake, score computation, dashboards, and DOCX/PPTX deliverables.
- **Priority-aware recommendations.** Led parallel workstreams across ingestion, data-collection UI, scoring, dashboards, deliverables, and practitioner workflows; modeled personas, themes, services, use cases, and leadership priorities to align recommendations with business goals.
- **Cross-stack integration and security.** Unified Java and Python services and project lifecycle workflows, ported Flask to FastAPI, moved secrets to AWS Secrets Manager, and drove CSDL, SonarQube, OAuth, CSP, and OWASP ZAP remediation.

Software Engineer IV – AppDynamics Customer Engineering

Feb 2022 – Sep 2022

- Architecture-level observability SME embedded with enterprise customers: mapped application architecture, defined KPIs with engineering leaders, and supported delivery including custom AppDynamics plugins.

Software Engineer III/IV – Collaboration Platform Engineering

Nov 2018 – Feb 2022

- Engineering lead on Cisco's Collaboration Workflow Manager: owned data-extraction and aggregation microservices, drove cluster consolidation, data migration, and rollouts for a Big Four U.S. bank and a global technology company; shipped a full-stack Webex Contact Center analytics dashboard (Angular, Spring Cloud, MongoDB).

Infosys

Aug 2011 – Nov 2018

Bangalore, India & San Francisco Bay Area

Full-Stack Engineer

Aug 2011 – Nov 2018

- Engineered the workflow-automation core of an enterprise provisioning platform for a global bank (jBPM directed-graph orchestration of Java services, Spring REST/SOAP APIs; 500K+ requests over five years); later technical lead driving the monolith-to-microservices re-architecture for a U.S. technology client.

Education

M.S. Computer Science (Artificial Intelligence), *Georgia Institute of Technology* — Atlanta, GA

Dec 2024

B.Tech. Electronics & Communications, *Punjabi University, Patiala* — Punjab, India

Jan 2011

Technical Skills

AI / ML — Systems & Retrieval: Agent SDK development, multi-agent orchestration, LangGraph state machines, MCP, ReAct, MCP tool-use benchmarking, LLM codegen with AST gating, hybrid search (dense + BM25 / TF-IDF), RRF, cross-encoder / LLM reranking, vector databases (Pinecone, pgvector, FAISS), hierarchical retrieval, NL-to-MongoDB / NL-to-SQL, GraphRAG, embeddings

AI / ML — Modeling, Evaluation & Safety: Fine-tuning SLMs / LLMs, LoRA / QLoRA, RAFT, synthetic data generation, RAG evaluation, LLM-as-judge, pairwise evaluation, RAGAS, agent harness design, calibration and refusal analysis, fault injection, AI guardrails, grounding / hallucination control, prompt-injection defense, tenant isolation

AI / ML Tools: PyTorch, Hugging Face Transformers, TRL, PEFT, vLLM, LangChain, LangSmith, Databricks, MLflow, Modal, pandas, NumPy, scikit-learn, XGBoost, TensorFlow, AWS Bedrock, AWS SageMaker, GPT-4o, Claude, Gemini, Llama, Azure OpenAI

Backend & Databases: Python (FastAPI / Flask), Java 21, Spring Boot 3 / Spring Cloud, Maven, REST / gRPC / SSE, microservices, distributed systems, event-driven architecture, system design, MongoDB Atlas, PostgreSQL, DuckDB, Apache Superset

Frontend: React, Angular 18, TypeScript, standalone components, signals

Security: HIPAA-aware design, CSDL compliance, OWASP ZAP, Trivy, Semgrep, Anchore, OAuth, JWT, AWS Secrets Manager, age-encrypted secret vault

Development & Cloud: Cursor, Claude Code, GitHub Copilot, Windsurf, Conda, Docker, Kubernetes, OpenShift, Terraform, CI/CD, Jenkins, AWS (S3 / ECR / Secrets Manager / OpenSearch / EC2 / IAM), GCP, Azure, Jira, Confluence, SharePoint, Splunk